MPS Capstone Project I (AEM 6991 LEC 004)

Dyson School of Applied Economics and Management · Fall 2026

Course

Fall 2026. Wednesdays, 2:00–4:30 pm. First meeting: August 26, 2026. Last meeting: December 2, 2026. Location: Riley-Robb Hall B15. Section: LEC 004.

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Office hours. Thursdays, 3:00–5:00 pm, 450B Warren Hall. Also by appointment.

Course website. arielortizbobea.github.io/aem6991 carries the schedule, slides, handouts and readings, with a page for every meeting. Canvas carries grades and submissions.

Credit hours. 3. Enrollment limited to AEM MPS students.

Course description

This is a core course for Master of Professional Studies (MPS) in Applied Economics and Management. It provides students with the opportunity to explore strategies for behavioral, quantitative, and qualitative problem-solving projects. We consider the conceptual challenges associated with identifying and defining a project topic and examine the practical tasks of selecting or collecting data, analyzing data, and reporting the results, including visualization and writing. The objective of the course is for students to understand how problems associated with different kinds of projects can be addressed with empirical methods. Many course activities will be structured as teamwork, and team leadership and management skills are a major component of the course. The course topics are introduced through readings, class discussion, and independent team-based research.

AEM 6991 is the first half of the MPS capstone sequence. In this section, each team moves from a vague interest to a research plan that can be executed in AEM 6992 in the spring. That plan is the deliverable.

The course runs in the order the work actually happens.

Part	What it covers
Getting started	Who is in the room, how the course runs, and what everyone is interested in
Foundations	Theory and causal graphs. Where research questions come from and how to filter them. Ethical and legal obligations, including IRB review and data-use agreements. Using AI in research
Methods & designs	Parameters, estimators and standard errors. How estimates fail, and what control variables can and cannot fix. Experiments, instrumental variables, regression discontinuity, difference-in-differences and panel designs
Communication	Writing, figures and speaking, each applied in session B to your own project

Part	What it covers
Delivery	The midterm, and the final presentations

Three commitments shape it.

The question comes before the data. Teams who find data first rationalise a question around whatever they found. That is the most common way a capstone goes wrong. The conceptual half is taught on examples from industry and the news, before anyone opens a dataset.

Finding the question is a skill. Most of your education has rewarded execution. AEM 6992 is where the executing happens. A bad question executed beautifully is still a bad project.

Writing and teamwork run through the whole term. Writing, figures and speaking each get a meeting. Each team writes its own ground rules in meeting 2, and every session B opens with a check-in that is written down and counts toward participation.

No prior econometrics or programming is assumed.

Audience and prerequisites

Open to, and required of, AEM MPS students.

There are no formal prerequisites. No econometrics, no prior programming, and none is taught. Where the course touches data directly, it does so through an AI assistant. No student is blocked by tooling.

If you arrive with statistics or coding experience, the data work will be easy. The design vocabulary and the verification habit will be new regardless of what you bring.

Learning outcomes

The registrar's outcomes for AEM 6991 are unchanged. By the end of the semester students will be able to:

1. **Teamwork.** Work effectively in small groups, complete tasks on time, and allocate work equitably among team members.
2. **Problem focus.** Select and define a project problem that demonstrates relevant knowledge and articulates the significance of the problem.
3. **Project design and workplan.** Develop a methodologically rigorous research design and identify key project tasks and outputs.
4. **Presentation of progress.** Write about, present, and discuss the project in class with analytical rigor.

This section adds two outcomes that serve the four above:

5. **Evidence sourcing.** Establish that the data a project needs exists, can be obtained on the project's timeline, and measures what the research question requires.
6. **Design vocabulary.** State what identification strategy a project relies on, what it assumes, and what it therefore can and cannot claim.

Assessment

Half the grade is individual and half is team-based.

Component	Type	Due	Weight
In-class quizzes	Individual	Most weeks through Nov 11, in class	15 %
Midterm	Individual	Nov 18, in class	15 %
Participation	Individual	Throughout	20 %
Individual subtotal			*50 %*
Project description — up to three pages, presented	Team	Oct 23; talk Oct 28	5 %
Proposal outline	Team	Nov 11	15 %
Final presentation	Team	Dec 2, in class	10 %
Research proposal, written	Team	Dec 11	20 %
Team subtotal			*50 %*
Total			**100 %**

Each component is scored out of 100 and weighted as above. Grading scale: A+ (≥ 98), A (≥ 92), A- (≥ 88), B+ (≥ 85), B (≥ 80), B- (≥ 75), C+ (≥ 70), C (≥ 65), C- (≥ 60), D (≥ 50), F (< 50). The final percentage is not rounded before the scale is applied.

In-class quizzes (15 %)

Short quizzes in class, on your own device. A quiz closes session A, and it is announced in advance together with the meetings it covers. Its questions come from those lectures. Expect one in most weeks through meeting 12. The exact number is not fixed.

Most are scenario questions. *Here is a situation — which design fits, and what would have to be true? Here is a design — what would break it? Here is a step an AI just did for you — how would you check it without reading the code?*

How the quiz grade is computed. Each quiz is scored as a percentage, and all quizzes count equally. Your lowest score is dropped and the rest are averaged. A missed quiz scores zero, and there are no make-ups. A quiz missed for an absence covered by an accommodation letter, a religious observance or documented illness is left out of the average.

Midterm (15 %)

In class on Wednesday, November 18. Open note, short answers. It is the whole meeting — there is no session B that week. The date is fixed and will not move.

You are given a situation and asked to design the study that would answer it. The question. The design, and the assumption carrying it. What data it requires. And what would undermine it.

It tests whether you can apply the design vocabulary to a situation you have not seen. There is nothing to revise beyond having turned up all term.

Participation (20 %)

Participation is scored out of 100. You start at 100 and lose points for the items below, down to a floor of zero. At a weight of 20 %, each point here is worth 0.2 of a point on the final grade.

	Points
Research interest survey not submitted, or late	−5
CITI human-participants training certificate not on Canvas by September 30, for a project with human participants	−5
Your team's session-B check-in not filed	−2 each
Peer evaluation not submitted on time	−3 each
Arriving late, or leaving at the break	−2 each
Absent without notice	−4 each

The research interest survey was due September 1 on Canvas. Teams were formed from it.

The session-B check-in is ten minutes at the start of every second half: what your team did since last week, what is stuck, and who does what next. Written down and handed in. It is the record that makes an uneven split of the work visible in week 4 rather than week 13.

Peer evaluation runs twice, at meetings 6 and 12, on CATME. It enters the grade only through the deduction above. Peer ratings do not adjust team grades.

Team deliverables (50 %)

One paper, reached in three milestones and presented once. A project description of up to three pages, then the whole argument as a detailed outline, then the written proposal, which the team presents at meeting 14 before handing it in. The argument is fixed before the sentences are, so a broken argument is caught at the outline where fixing it costs an afternoon rather than a week.

#	Deliverable	What it is	Weight	Due
1	**Project description**	Up to three pages, due on Canvas the Friday after meeting 9, then three minutes in front of the room at meeting 10. The question and who acts on the answer, why it matters, how you would answer it, the data and how you get it, what could go wrong, and who on the team does what. This is the first time the project has to survive an audience, and the pages are what you hand in. The brief is on the project description page.	5 %	Oct 23 (week 9)
2	**Proposal outline**	The whole proposal as a detailed outline. One topic sentence per paragraph, in order, with the evidence noted	15 %	Meeting 12 — Nov 11 (week 12)

#	Deliverable	What it is	Weight	Due
		underneath — the citation, the number, the figure you will point at. It has to cover the question, the identification strategy and the assumption carrying it, the unit and sample, the data and exactly how you get it, the threats to validity, and the work plan for the spring. Exchanged with a partner team at meeting 12, who referee it. If the argument does not work here, it will not work at 5,000 words.		
3	**Final presentation**	The proposal presented to the class in the time allowed, followed by questions from the room.	10 %	Meeting 14 — Dec 2 (week 15), in class
4	**Research proposal**	~5 pages, written out from the outline you workshopped and the presentation you gave. The plan somebody could pick up and execute in January.	20 %	Dec 11 (week 16)

Every member of a team receives the same grade on each team deliverable.

The data section of the outline is the one that kills projects. The most common way an MPS capstone fails is not a bad question. It is a good question whose data never arrives. So the outline makes you name the dataset, its holder, the access route, the timeline and your plan B.

How the proposal gets built

You do not write the proposal in November. You accumulate it. Session A introduces a concept; session B makes you use it, first on a shared case and then on your own project. That second half is what you hand in before you leave, and it is a piece of your proposal.

Meeting	Session B leaves you with
1 Introductions	Your team's question sheet: candidate questions, each run through the four tests of a good idea, with the ones that fail killed on the record and the reason named
2 Theory and causality	Your team's ground rules, and the first causal graph of your own question — treatment, outcome, the back door you must close, and the variable you must not touch

Meeting	Session B leaves you with
3 Finding the question	Your team's question sheet, three candidates each labelled with its route, its shape and who acts or pays, run through the four filters with one killed on the record, and the resource map of the survivor with the row that stops them dated and the question it becomes if that row does not close
4 Ethical and legal obligations	Your project's IRB and DUA checklist, signed by every member: whether it needs an IRB application, whether any data source needs a data-use agreement, and the date the slowest gate has to start
5 Using AI in research	Your candidate dataset interrogated — or, if you do not hold one yet, the named access route and plan B for getting it
6 Models and uncertainty	The overclaim your own project is most at risk of making
7 Model failures	The one threat your design must survive — its direction, and whether controls can close it
8 Research designs I	Whether an experiment, instrument or discontinuity is open to you
9 Research designs II	Your design defended in public, plus the attack that landed — the identification section and the limitation
10 Writing	Your problem statement, as an outline somebody outside your team could follow
11 Figures	Your project's key figure, drawn twice — as you expect it to look, and as it looks if you are wrong
12 Speaking	Your final-presentation deck, rehearsed and attacked — plus the gap between the claim you made and the claim the room heard
14 Final presentations	Your plan, pitched — and what your team owes AEM 6992 over the break

These weekly pieces, with the project description and the outline, are the checkpoints that keep each team on track for the proposal.

What you are graded against

Across AEM 6991 sections, the criteria for written team work are the **required section headings**. If a heading has nothing under it, that is the finding.

The outline and the proposal must both cover, under these headings: research problem · practical justification · what is already known · conceptual model · methodological approach · study sample · **data and access** · internal validity · external validity · **limitations and self-critique** · work plan · research ethics · references.

Excellent	Every heading is answered from your own project. The claim is in the first sentence. The evidence supports it. Somebody outside your team could act on it.
Adequate	The headings are answered, but some are generic — true of any project, not this one.

Weak	Substantial sections restate the prompt or borrow text. The argument does not cohere.
Unacceptable	Not submitted, submitted beyond the late policy, or so thin it shows no thought.

The final presentation is graded on the same scale. It is also graded on whether it fits the time allowed and on how the team answers questions.

Two questions move a grade more than anything else. **Is the claim in the first sentence?** And **could it come out the other way?**

Late policy. Written team deliverables lose 10 % up to 8 hours late, 50 % from 8 to 72 hours, and are not accepted after 72 hours. Notify me before the deadline if something has gone wrong.

Course schedule

Fourteen meetings. Each runs the same shape:

Time	
2:00–3:15	Session A — the concept, on worked examples
3:15–3:30	Break
3:30–4:30	Session B — **your own project**, submitted before you leave

When there is a quiz, it closes session A.

Session B tests the day's concept on a shared case, then applies it to the team's own project. Early in the term that means drawing your graph, naming your design and working out what data it needs. Later it means writing, figures and a rehearsed presentation of your project. The output is submitted the same day. Accumulated, these pieces are most of the research proposal.

Every meeting has its own page on the course site, carrying the readings, slides, and materials for that day.

#	Date	Meeting	Readings	Due
		Part 1 — Getting started		
1	Aug 26	**Introductions**	Adams ch. 1	Research interest survey, on Canvas before meeting 2
		Part 2 — Foundations		
2	Sep 2	**Theory and causality**	Remix ch. 3–4; Mostly Harmless ch. 1	Your team's written ground rules — agreed and handed in today
3	Sep 9	**Finding the question**	Adams ch. 3–4	
4	Sep 16	**Ethical and legal obligations**	Adams ch. 2	CITI human-participants training, for teams whose project involves human participants —

#	Date	Meeting	Readings	Due
				certificate on Canvas before meeting 6 (September 30)
5	Sep 23	**Using AI in research**	Adams ch. 7	
		Part 3 — Methods		
6	Sep 30	**Models and uncertainty**	Adams ch. 11–12	Rate your teammates
7	Oct 7	**Model failures**	Remix ch. 5; Adams ch. 14	
8	Oct 14	**Research designs I**	Remix ch. 4, 6–7; Mostly Harmless ch. 2, 4, 6	
9	Oct 21 · Zoom	**Research designs II**	Remix ch. 8–11; The Effect ch. 16–18; Mostly Harmless ch. 5	Project description — up to three pages, on Canvas by Friday, October 23
		Part 4 — Communication		
10	Oct 28	**Writing**	Adams ch. 4, 16	Project talk — three minutes, in session B
11	Nov 4	**Figures**	Zelazny; Adams ch. 11	
12	Nov 11	**Speaking**	Adams ch. 16	Proposal outline · Rate your teammates
		Part 5 — Delivery		
13	Nov 18*	**Midterm**		
	Nov 25	*Thanksgiving recess — no class*		
14	Dec 2	**Final presentations**	Adams ch. 15	Final presentation, in class
	Dec 11	*Research proposal due — the final write-up. Date to be confirmed*		

* These dates are not final: **Nov 18** — Instructor off campus — the midterm is proctored; arrangements announced by week 10

Thanksgiving recess falls on Wednesday, November 25 — no class. The research proposal is due December 11, to be confirmed against the exam schedule.

Readings and handbooks

There is no book to buy. Everything is free, either openly online or through the Cornell library.

Quizzes are drawn from the lecture, not from a reading, so nothing below is a gate. These are the shelf you reach for when a session raises something you want more of, and each meeting page points at the chapters that fit it.

The three handbooks

Cunningham, *Causal Inference: The Remix* — the second edition of *The Mixtape*, free to read online. The first edition is still up. This is the main one for us: readable without prior econometrics, and it covers the whole inference half.

1–2	Foundational ideas · which causal inference?
3–4	**Directed acyclic graphs** · potential outcomes and randomization
5	**Unconfoundedness**
6–7	**Regression discontinuity** · **instrumental variables**
8–11	Causal panel designs · **difference-in-differences** · complex DiD · synthetic control

Adams, Khan, Raeside & White (2007), *Research Methods for Graduate Business and Social Science Students* (SAGE) — the shared text across AEM 6991 sections, free through the Cornell library. Sign in with your NetID. It is where the non-inference craft lives: the research cycle, literature reviewing, sampling, surveys, and writing up.

1–2	Introduction to research · methodology and **research ethics**
3–4	The research cycle · **literature review and critical reading**
5	**Sampling**, and sample size determination
6–9	Primary data · **secondary data** · **surveys** · interviews
10–13	Qualitative analysis · descriptive statistics · correlation and regression · advanced methods
14	**Tests of measurement and quality** — reliability, validity, generalisability
15–16	Conducting your research · **writing and presenting**

Angrist & Pischke, *Mostly Harmless Econometrics* (Princeton, 2009) — the standard graduate reference for the designs in the Methods & designs part of the course.

> ⚠ Warning
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> **This one assumes econometrics.** It is not assigned and you are not expected to read it. It is here because it is the book the designs in this course come from, and because a few of you will want it — or will meet it in a job. Chapter 1, *Questions about Questions*, is eight pages and needs no maths. Start there or nowhere.

1–2	**Questions about questions** · the experimental ideal
3	Making regression make sense
4–6	Instrumental variables · fixed effects, DiD and panel data · regression discontinuity

7–8 Quantile regression · standard error issues

Also recurring

Huntington-Klein, *The Effect*, free online and the gentlest of the four. Sandberg & Alvesson on problematization; Haas & Mortensen on teams; Stanley & Castles, *The So What Strategy*; Zelazny, *Say It With Charts*.

Course policies

Attendance

The format is once a week. So missing one meeting means missing two sessions.

Attendance is recorded by the session-B check-in, which cannot be filed retrospectively. To give notice of an absence, email me before the meeting starts. An absence with notice costs no participation points.

A quiz missed through absence scores zero, unless the absence is excused as set out under the quizzes. The dropped score absorbs one absence without penalty and without a conversation.

If you are going to miss more than two meetings, talk to me early rather than late.

AI policy

Generative AI tools are **permitted by default** for all coursework. Students are expected to use them, and to use them well. Three rules apply to every deliverable:

1. **Disclose what you used.** Every submitted deliverable carries a short disclosure block: which tools, on which parts of the work. The form is brief; what matters is that it exists.
2. **Verify what the AI produced.** AI output is your work, exactly as code from a colleague would be. Two failure modes are graded against explicitly, because they are the ones that destroy a research proposal:
   - **Fabricated citations.** Models invent references, and they do it convincingly. Every reference in every submission must be one you have opened. A citation to a paper that does not exist is treated as fabrication.
   - **Plausible-but-wrong design advice.** Models will confidently recommend an estimator whose assumptions your data violates. The design you propose is yours.
3. **Do not paste restricted data into a public chatbot.** Confidential, IRB-restricted, and proprietary data stay on your machine. Cornell's rule gives restricted a definition: nothing classified as moderate risk or above goes into a publicly available AI tool, and licensed data counts. The approved tools are listed at the AI Innovation Hub.

Failure to disclose is an academic integrity violation under the Cornell Code. Failure to verify is graded against the rubric like any other error.

> **i** Note
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> **Note.** Other sections of AEM 6991 restrict generative AI. This one takes the opposite position, deliberately.
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> The deliverable here is a research proposal. That is exactly the genre where unverified AI output does the most damage. So the skill worth teaching is verification, not abstinence.

Academic integrity

Every student is expected to abide by the Cornell University Code of Academic Integrity. Work submitted for credit is your own.

AI assistance is permitted under the policy above. **Failure to disclose AI use is a violation.** So is submitting a fabricated citation as a real one.

Collaboration is expected on team assignments. It is not permitted on individual ones.

Accommodations

If you need academic accommodations, contact Student Disability Services. Then give me your accommodation letter as early in the semester as you can, so arrangements can be made.

Inclusion

Students in this course come from many countries and backgrounds. Everyone should be able to ask a question and be wrong out loud. Argue hard about ideas and treat the people holding them with respect. If something in class gets in the way of that, tell me or the TA.

Data practices

Quiz submissions are tied to your netid and timestamped. They are used for grading and attendance.

The roster, your submissions, and any per-student credentials are held in Cornell-managed or instructor-controlled storage. None of it is published. Course materials in the public repository never contain student-identifying data.

Last updated: 2026-10-19. Dates marked with an asterisk in the schedule are still to be confirmed.